Computer Science Portfolio Project Report

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Github Repository Link: https://github.com/arjunbijoy1917r-cmyk/computer-science-lab.git

Portfolio link : https://arjunbijoy1917r-cmyk.github.io/computer-science-lab/portfolio_site/

Contents

1 Introduction

This report explains the content of my computer science portfolio website. The aim of the portfolio showcase the talents and skills I achieved as a Software Engineering student and including the projects I have developed. The portfolio is built in a simple design and it is developed using HTML and CSS. It contains my personal information, my achievements, my completed work, and technical skills.

The website is also linked to my GitHub so that visitors can see the source code and project repositories. The portfolio webpage is hosted using GitHub Pages. The GitHub repository stores the website files, including the HTML page, CSS stylesheet, and asset folder. Overall, The report shows the exact tools and skills I used to develop this portfolio.

2 Website Structure

The portfolio website is a simple personal webpage. The main navigation bar at the top includes links to Home, About, Achievements, Work, Skills, and Contact Me. The simple structure was selected because it allows visitors to quickly move to the information they need.

2.1 Home Section

The Home section starts with greeting ,short introduction and my name. Also including a short intro that I am a Software Engineering student. This section is the first impression of my portfolio and it makes the website a personal feel.

2.2 About Section

This section explains who I am and what I am currently studying. It mentions my interest in programming, web development, and databases. Also the about explains that I am still building my skills and learning tools such as HTML, CSS, Python, Java, SQL and LaTeX.

2.3 Achievements Section

The Achievements section has the progress I have made so far and it also includes that I have completed Python and SQL database projects, technical exercises, and beginner-level projects. I included this section to show that the portfolio is not only about my final works but also the progress of my learning.

2.4 Work Section

The Work section is one of the most important parts of the website. As it presents my two main projects:

- **Python University System:** A Python project based on a university system. It has the concepts such as classes, objects, students, courses, instructors, and enrollments.
- **University Database Project:** This is a SQL database project based on a university management system. It has tables, sample data, queries, relationships, and database design practice.

Each project has a button that opens the related GitHub project. This makes it easy to navigate and explore the GitHub repositories.

2.5 Skills Section

The Skills section shows the main parts, technologies and areas I am currently learning. This include basic Python coding, Java, HTML, CSS, SQL Database, and LaTeX. I kept this section short because it gives the quick overview of my current technical skills.

2.6 Contact Section

The Contact Me section includes my email address. This section is very useful because people can contact me for queries and updates portfolio. Also my other contact details are included along with the CV.

3 GitHub Repository Structure

The current GitHub repository is named as computer-science-lab. In this repository, the portfolio website is stored in portfolio site folder. The portfolio site folder contains the main website files. The index.html file contains the main webpage content and structure. The styles.css file controls the design, layout, colours, spacing, and hover effects. Assets is the folder used to keep the media files including images. The file named projects.html is used to store project contents and it can be developed into a project page if needed.

This repository structure is useful because it separates the portfolio website from other files in the repository. It also makes the website easier to manage because if I need to edit or make improvements in the design and I can edit the CSS file as well. If I need to update the content, I can edit the HTML file also.

4 Design Decisions

The design of the portfolio is made simple, readable, and suitable for people to easily browse my computer science portfolio. I avoided making the website too complicated because the main purpose is to present my work clearly.

4.1 Simple Navigation

A navigation bar at top was used so that visitors can easily move between the main sections of the page. The navigation links use sections like About, achievements, Work, Skills, and Contact me. These names are easy to understand and easy to browse through the portfolio.

4.2 Colour Choice

In the website I have used a light background with white sections and a brownish orange accent colour. This colour is used in headings, borders and buttons. The light background improves readability, while the accent colour gives the website a more personal and professional style with my profile picture.

4.3 Section-Based Layout

Each main part of the portfolio is placed in a separate section. It will make reading easy since information will not be mixed up on the site. The section cards also helps to browse the page and gives a clean structure and help visitors to read the contents quickly.

4.4 Project Cards and Buttons

The projects are displayed using project boxes. Each project has a title, description, and button linking to GitHub. This part was important for me because this the proof of my learning. The buttons make the website more interactive and help users redirect or open the project repositories directly.

4.5 Beginner-Friendly Design

The website was designed with HTML and CSS. This was intentional because it reflects my current learning stage. Using simple code also helps me understand each part of the website properly. I have also referred many videos that was published on the canvas page, which makes it easier for me to develop the webpage. The portfolio is not only my final product, but also this shows parts of my learning process.

5 Tools and Technologies Used

This portfolio was created using the following tools and technologies:

- **HTML:** is used to create the structure and content of the webpage, including headings, paragraphs, sections, links, and buttons.
- **CSS:** is used to style the website, by including colours, fonts, spacing, layout, borders, shadows and some hover effects.
- **GitHub:** used to store the project files and manage the portfolio repository.
- **GitHub Pages:** is used to host the portfolio website online so that it can be accessed through a public URL.
- **LaTeX:** Used to prepare this project report in a structured and professional format.
- **Web Browser:** is used to test how the portfolio looks after uploading it to GitHub Pages.

These tools were selected because they are suitable for a basic computer science portfolio. HTML and CSS are the core background technology for building the webpage. GitHub and GitHub Pages are useful because they allow the website to be stored and published online for free.

6 Implementation Explanation

The implementation of the portfolio started with creating the basic HTML structure. I divided the page into different sections so that each part of the portfolio can have a clear aim.The website is easier to use because the visitor does not need to scroll manually through the whole page.

After creating the HTML, I used CSS to improve the overall appearance of the website. The CSS file controls the top menu, greeting area, image placement, section cards, project boxes, buttons, and contact links. The use of a separate CSS file was a good decision for me because it keeps the design separate from the HTML content.

The website was then uploaded to the GitHub repository inside the portfolio _site folder. After uploading the files, GitHub Pages was used to publish the website. The final webpage can be accessed using the GitHub Pages URL. This process helped me understand how websites can be hosted from a repository because I was confused when my professor showed me his GitHub portfolio page and how folder paths affect the final website link.

7 Portfolio Content

The content of the portfolio is based on my current learning and projects. I included information that is relevant to my studies and technical development. The About section explains my background as a Software Engineering student. The Achievements section shows my progress in completing projects and some exercises I have done. The Work section shows my Python and SQL projects because both of these are important examples of my practical learning.

The Python University System project shows my understanding of basic programming and object oriented concepts. The University Database Project shows my understanding of database tables, relationships, queries, and database design and also helped me in analysing the MariaDB and MySQL. These projects are suitable for the portfolio because they are connected to computer science and software engineering field.

8 Strengths of the Portfolio

One strength of the portfolio is that it has a simple design and easy to understand. The sections are clearly labelled and also the navigation menu helps users move around the page. Another strength is that the project links connect directly to GitHub, which allows visitors to check the actual project work.

The design is also consistent because the same accent colour is used throughout the website. The use of separate HTML and CSS files makes the project easier to edit and impro and browse. Honestly the portfolio shows my current learning stage, which is useful for a student portfolio.

9 Weaknesses of the Portfolio

The portfolio also has some weaknesses. The design is currently basic and does not included any advanced features such as animations, responsive designs improvements, or a working contact form. I managed to make the website myself so that it doesn't showcase some other works. Some content can also be elaborated, especially the project descriptions.

Another weakness I felt is that the website only has text and my cover picture. Adding more visual elements like screenshots, diagrams, or short descriptions of the development process, may make the portfolio more professional and complicated.

10 Future Improvements

In the future , I would like to improve the portfolio by adding more features and better content which I can develop myself and have some planned improvements like:

- Adding screenshots of each project so visitors can see how the project will be looking like.
- Adding a separate project details page for each project.
- Improving the mobile view so the website looks better on phones and tablets.
- By adding a working contact form.
- Adding my CV as a downloadable PDF.

These improvements can make my portfolio more professional and interactive. They would also help to show case my progress over time as I gain more knowledge and technical skills.

11 Conclusion

In overall conclusion, this portfolio project helped me practise basic web development and learn how to publish a website using GitHub Pages. The website presents my background, achievements, projects, skills, and contact information. The GitHub repository also shows the structure of the website and allows the source code to be reviewed.

Although my portfolio is a simple and basic design, it is a good starting point for showing my computer science learning progress. The main strengths are its clear structure, simple navigation, project links, and consistent design. The main areas for improvement are responsive design and some additional interactive features which are lacking. Overall, this project helped me capture the connection between designing a website, content organisation, and online hosting.